

Module -1

❖ Session:1

Q.1. Name 5 IT job roles and write one line about what each role does.

Answer:

1. **Software Developer** – Develops and maintains software applications and websites.
 2. **Web Developer** – Creates and manages websites using frontend and backend technologies.
 3. **Data Analyst** – Analyzes data to find useful information and support business decisions.
 4. **Data Scientist**: Uses data to solve problems and make predictions.
 5. **Software Tester** – Tests software to find bugs and ensure it works correctly.
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Q.2. List 3 product companies and 3 service companies with one example of work each does.

Answer:

Product Companies:

1. **Microsoft** – Develops products such as Windows and Microsoft 365.
2. **Google** – Develops products such as Google Search, Gmail, and Android.
3. **Adobe** – Develops software products such as Photoshop.

Service Companies:

1. **TCS** – Provides IT consulting and software development services to clients.

2. **Infosys** – Provides IT consulting, application development, and business solutions.
 3. **Wipro** – Provides IT services such as cloud computing and software development.
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Q.3. Full forms and one-line use.

Answer:

- **GitHub** – Used to store and share code.
 - **Slack – Searchable Log of All Conversation and Knowledge** – A communication platform used by teams for messaging and collaboration.
 - **Jira – Jira is not an acronym** – It is a project management and issue-tracking tool used to manage tasks and bugs.
 - **IDE – Integrated Development Environment** – Software that provides tools for writing, running, and debugging code.
 - **API – Application Programming Interface** – Allows different software applications to communicate and exchange data.
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Q.4. Name any 5 programming languages and mention one popular use case.

Answer:

Programming Language	Popular Use Case
Python	AI, Machine Learning, and Backend Development
Java	Enterprise and Android applications
JavaScript	Interactive websites and web applications
C++	Game development and system software

Programming Language	Popular Use Case
C#	Windows applications and game development

Q.5. Difference between a developer and a tester in 2 lines.

Answer: =>A **developer** writes and maintains code to build software features and applications.

=>A **tester** checks the software to find bugs and ensure that it works according to requirements.

❖ Session:2

Q.1.Three Main Layers of Software Architecture.

Answer:

1. **Presentation Layer** – It is the part that the user can see and use.
 2. **Business Logic Layer** – It contains the main rules and logic of the application.
 3. **Data Layer** – It stores, retrieves, and manages data in the database.
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Q.2.Client and Server.

Answer:

- **Client** – A client is the device or app that asks for information.

- **Server** – A server gives the information to the client.

Real-life example: When we open YouTube on our mobile, the **mobile is the client** and the **YouTube server** sends the videos and data to it.

Q.3. Desktop Apps, Web Apps, and Mobile Apps.

Answer:

Type	Difference 1	Difference 2
Desktop App	Installed on a computer.	Usually works on Windows, Mac, or Linux.
Web App	Runs in a web browser.	Usually does not need to be installed.
Mobile App	Designed for smartphones or tablets.	Usually installed from an app store.

Q.4. Three Databases

Answer:

1. **MySQL** – SQL Database
 2. **PostgreSQL** – SQL Database
 3. **MongoDB** – NoSQL Database
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❖ Session:3

Q.1.Full Form and Use of HTTP, HTTPS, FTP, SMTP.

Answer:

1. **HTTP** – Hyper Text Transfer Protocol – It is used to transfer data between a browser and a web server.
 2. **HTTPS** – Hyper Text Transfer Protocol Secure – It is used to transfer data securely over the internet.
 3. **FTP** – File Transfer Protocol – It is used to transfer files between computers.
 4. **SMTP** – Simple Mail Transfer Protocol – It is used to send emails.
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Q.2. Difference between IPv4 and IPv6.

Answer:

IPv4 uses a 32-bit address and has a limited number of IP addresses.

IPv6 uses a 128-bit address and provides many more IP addresses.

Q.3. Run nslookup google.com and write the IP address.

Answer:

Command: nslookup google.com

IP Address: 172.217.26.

Q.4. Meaning of HTTP Status Codes.

Answer:

1. **200** – The request was successful.
 2. **301** – The page has been moved permanently.
 3. **404** – The page was not found.
 4. **500** – There is a server error.
 5. **403** – Access to the page is forbidden.
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Q.5. Explain DNS Hierarchy www.google.com

Answer:

1. **Root (.)** – It is the top level of the DNS hierarchy.
2. **TLD (.com)** – It shows the top-level domain.
3. **Domain (google)** – It is the main name of the website.
4. **Subdomain (www)** – It is a part of the main domain.

Example: www.google.com → **www** = Subdomain, **google** = Domain, **.com** = Root.

❖ Session:4

Q.1. List all 6 phases of SDLC in order with one line about each.

Answer:

1. **Planning** – In this phase, the project requirements and goals are decided.
2. **Analysis** – In this phase, the requirements of the users are studied.
3. **Design** – In this phase, the structure and design of the software are prepared.
4. **Implementation** – In this phase, the actual code of the software is written.

5. **Testing and Integration**– In this phase, the software is tested to find and fix bugs.
 6. **Maintenance** – In this phase, the software is updated and maintained after release.
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Q.2. Write 3 differences between Agile and Waterfall models.

Answer:

1. **Agile** works in small steps, while **Waterfall** works step by step in a fixed order.
 2. **Agile** allows changes during development, while **Waterfall** is difficult to change after a phase is completed.
 3. **Agile** delivers software in small parts, while **Waterfall** usually delivers the complete software at the end.
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Q.3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Answer:

1. **Product Owner** – Manages the product requirements and decides what should be built.
 2. **Scrum Master** – Helps the team follow Scrum rules and removes problems.
 3. **Developers** – Build and test the product according to the requirements.
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Q.4. What is a Sprint in Scrum? Mention its typical duration.

Answer:

A **Sprint** is a fixed period in which the Scrum team works to complete a set of tasks.

Its typical duration is **2 to 4 weeks**.

Q.5. What do these Jira board columns mean?

Answer:

1. **To Do** – Tasks that have not been started yet.
 2. **In Progress** – Tasks that are currently being worked on.
 3. **Done** – Tasks that have been completed.
 4. **Backlog** – A list of tasks or requirements that are planned for future work.
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❖ Session:5

Q.1. Write the Git command for each.

Answer:

1. **Initialize a repository:**
`git init`
 2. **Stage all files:**
`git add`
 3. **Commit with a message:**
`git commit -m "First commit"`
 4. **Push to remote:**
`git push origin main`
 5. **Pull latest changes:**
`git pull origin main`
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Q.2. Difference between git fetch and git pull.

Answer:

git fetch downloads the latest changes but does not merge them into your current branch.

git pull downloads the latest changes and also merges them into your current branch.

Q.3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Answer:

Repo Link: [ayushikhunt/Git-hub-test](https://github.com/ayushikhunt/Git-hub-test)

Q.4. Difference between Fork and Pull Request.

Answer:

Fork creates a copy of someone else's repository in your own GitHub account.

Pull Request is a request to add your changes to the original repository.

Q.5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch.

Answer:

```
git branch feature-test
```

```
git checkout feature-test
```

```
git add .
```

```
git commit -m "Add feature test"
```

```
git checkout main
```

```
git merge feature-test
```